



Mowthooq—Smart مَوْتُوق Verification & Authoring for Content Creators

Intelligent tools for content creators to verify, correct, and
enhance their narratives.

The Challenge

The digital age brings a deluge of information, often making it difficult to discern truth from falsehood. Content creators face increasing pressure to produce accurate, polished, and accessible material while battling the spread of misinformation and errors.

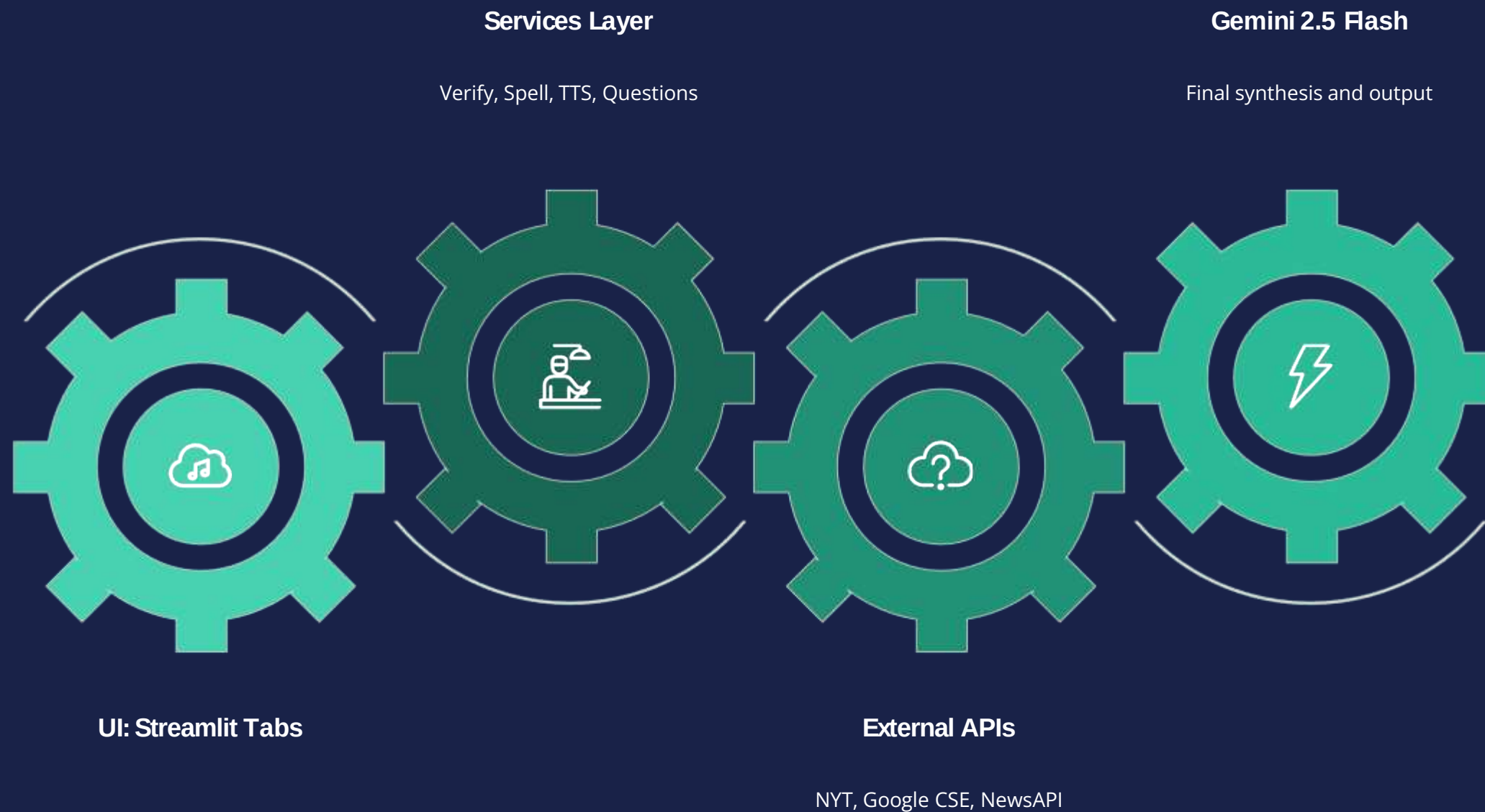


Our Solution

Mowthooq provides a suite of intelligent tools designed specifically for content creators. From verifying news and correcting text to enhancing accessibility and generating insightful questions, Mowthooq empowers users to create high-quality, trustworthy content with confidence.



Mowthooq Architecture



Core AI Components



VeriNews: AI-Powered Verification

Leverages Gemini 2.5 Flash to assess claims and return a detailed JSON output, including result, confidence, explanation, and sources.



Intelligent Retrieval

Integrates with NYTimes API and Google CSE (with optional NewsAPI) to gather comprehensive supporting or contradicting sources.



SpellGuard: Multilingual Correction

Combines English SpellChecker with AraBERT (fill-mask) for precise correction of both English and Arabic text.



TTS: Accessibility through Audio

Utilizes ChatterboxTTS (with CUDA support) to instantly convert summary text into high-quality audio files.



Dynamic Question Generation

Uses Gemini to generate concise, relevant questions based on user-provided topics, fostering deeper understanding.

VeriNews: The Logic of Trust

Mowthooq's VeriNews module processes claims and provides a comprehensive JSON output. This includes:

```
{  "result": "True/False/Unclear",
  "confidence": "0-100",  "explanation": "...",
  "trusted_sources": [],  "category": "...",
  "stance": "...",  "sentiment": "..."}

```

Confidence Thresholds

- If "result" is True and "confidence" is ≥ 70 , the claim is categorized as **True**.
- If "result" is Unclear and "confidence" is ≥ 70 , it's categorized as **Partially True**.

Trusted Sources Filtering

Mowthooq prioritizes information from an approved list of reliable domains, ensuring high-quality source verification.

bbc.com, cnn.com, reuters.com, nytimes.com,
aljazeera.net, ap.org (and many more)

SpellGuard: Precision Correction

Correction Pipeline

Text is segmented, then processed by specialized correctors:

- **English text:** Handled by a dedicated SpellChecker.
- **Arabic text:** Corrected using AraBERT's fill-mask capability for a limited number of words (default 10) to maintain performance.

Output & Quality Assurance

Mowthooq provides the corrected text along with a clear explanation of all changes made. Each correction comes with a quality badge, assuring the user of the accuracy and reliability of the suggested edits.

Accessibility: Text-to-Speech (TTS)

With a single click, convert your summarized text into an instant WAV audio file. This feature significantly enhances accessibility for visually impaired users.

 **GPU Consideration:** Please note that simultaneous heavy use of TTS and AraBERT on the same GPU might lead to performance warnings.



Thought-Provoking Questions

Mowthooq's intelligent question generation module helps content creators craft engaging and insightful questions to deepen audience understanding.

Dynamic Prompting

A built-in prompt, "**Generate N concise, thought-provoking questions about:**", guides Gemini to create relevant queries.

Customization Options

Users can specify the language (Arabic/English) and the number of questions (3-20) to perfectly suit their needs.

Example: AI in Education

- How can AI personalize learning experiences for diverse student needs?
- What ethical considerations arise when integrating AI into educational assessment?
- How might AI tools empower educators and reduce administrative burdens?
- What future skills will be essential for students in an AI-driven educational landscape?
- How can AI ensure equitable access to quality education globally?

Performance Metrics

Mowthooq is engineered for efficiency and accuracy, with key performance indicators ensuring a seamless user experience.

<3s

Response Time

Target response time for each module during demonstrations, ensuring a smooth and responsive user experience.

>3

VeriNews Sources

Minimum number of trusted or conflicting sources identified by VeriNews for a robust verification.

显著

SpellGuard Error Reduction

Significant reduction in common errors on a sample text, ensuring high text quality.

<1.5s

TTS Audio Generation

Time to generate an audio file for a short summary, ensuring quick accessibility.

These metrics drive our commitment to delivering a powerful and reliable tool for content creators.

The Impact: Data-Driven Insights



Combating Misinformation

59% of people are concerned about what is real or fake online when it comes to news. Mowthooq helps navigate this uncertainty.

[Source: Pew Research Center, "Social Media and News Fact Sheet"]



Eliminating Errors

Spelling errors can drop text credibility by **~8.9 points** out of 100. **59% of users** won't engage with content that has obvious typographical errors.

[Source: Grammarly, "The Cost of Poor Writing"; Econsultancy, "Why Bad Grammar Costs Businesses Billions"]

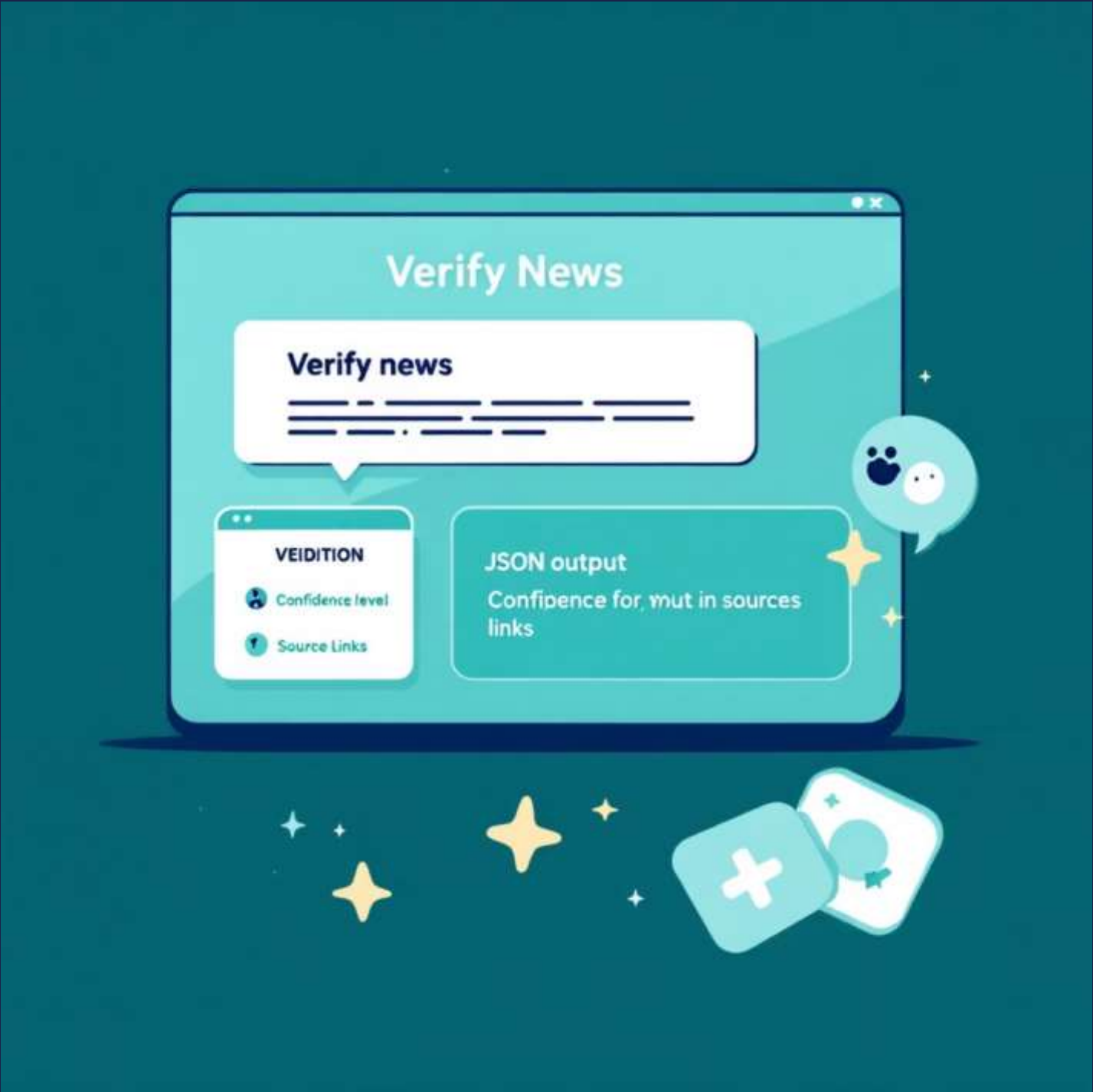


Enhancing Accessibility

At least **2.2 billion people** have near or distance vision impairment, and **430 million** have disabling hearing loss. Mowthooq ensures content reaches everyone.

[Source: WHO, "Blindness and vision impairment"; WHO, "Deafness and hearing loss"]

Mowthooq in Action: Interface Walkthrough



Verify News: Input text or a link, then instantly see the verdict, confidence score, and detailed sources supporting or refuting the claim.



Recent Results: Access a comprehensive history of all your verification operations, complete with links to the original sources for easy reference.